



LEAD MATRIX
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FUNCTIONAL SPECIFICATION DOCUMENT

ARA Lead Matrix CRM Platform

Functional Behaviour, Screen & API Specification — Full System

A complete functional specification covering every module, screen, field, validation, workflow, state transition, business rule, API behaviour, message, error path and integration across the agency console and the client self-service portal.

PREPARED FOR
Product, Engineering & QA

DATE OF ISSUE
17 July 2026

PREPARED BY
Functional Consulting Office

CLASSIFICATION
Confidential

VERSION
1.0 (Baseline)

COMPANION TO
ARA-BRD-CRM-2026-001

Ownership, governance and distribution controls for this Functional Specification Document (FSD).

Attribute	Detail
Document Title	ARA Lead Matrix CRM — Functional Specification Document
Document Reference	ARA-FSD-CRM-2026-001
System Name	ARA Lead Matrix CRM (internally "New Ara CRM")
Companion Document	Business Requirements Document — ARA-BRD-CRM-2026-001
Version / Status	1.0 — Baseline — Approved for Review
Classification	Confidential
Date of Issue	17 July 2026
Technology Baseline	Node.js/Express (ES modules), MongoDB (Azure Cosmos DB), React SPA + Redux Toolkit + MUI, JWT auth

Purpose & Audience

This FSD translates the business requirements into precise, testable functional behaviour. It is written for the product owner, engineering and QA teams, and technical stakeholders who need to understand exactly how each screen, field, workflow, API and integration behaves — including validations, messages, error paths and edge cases.

Approval & Sign-Off

Role	Responsibility	Signature	Date
Product Owner	Confirm functional completeness		
Engineering Lead	Confirm technical accuracy		
QA Lead	Confirm testability of criteria		
Business Analyst	Confirm alignment with the BRD		

Basis. This specification was produced through a complete source-level review of the delivered system — data models, API routes and controllers, middleware, background jobs, integrations, and the React front-end — and reflects the system *as built and in production*.

Revision History

Version	Date	Author	Summary
0.1	11 Jul 2026	Functional Consulting Office	Section outline and screen-inventory skeleton.
0.6	15 Jul 2026	Functional Consulting Office	Screen specifications, field definitions, validations and API behaviour drafted from source.
0.9	16 Jul 2026	Functional Consulting Office	Workflows, state transitions, messages, error/edge cases and acceptance criteria added.
1.0	17 Jul 2026	Functional Consulting Office	Baseline release — full 36-section FSD approved for circulation.

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Functional Overview

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ARA Lead Matrix is a multi-tenant performance-marketing CRM that connects ad spend, lead generation, telecalling and results into one live system, exposed through two authenticated experiences.

Functionally, the platform performs five continuous jobs:

Function	Behaviour
Acquire	Automatically ingests campaigns, ad sets, ads, daily insights, lead forms and leads from Meta and Google Ads on a recurring schedule; Meta leads additionally arrive in real time via signed webhooks.
Attribute & Reconcile	Normalises each lead to a source/platform, keeps counts consistent via a single metrics service, and reconciles every ad-spend rupee into an immutable billing ledger.
Work	Presents leads on a validated 26-column telecalling sheet where users log calls, responses, appointments and follow-ups, with automation for reminders, history and dormancy.
Report	Renders live dashboards, EOD and Monthly Abstract reports, comparisons and analytics — auto-refreshing and identical across every surface.
Govern	Enforces two role systems, strict tenant isolation, encrypted secrets, and full audit/observability of syncs, raw leads and financial entries.

25

FUNCTIONAL
MODULES

2

AUTH REALMS

18

API ROUTE GROUPS

24

DATA ENTITIES

~30s

LIVE REFRESH

5.1 Functional principles enforced by the system

- **Single source of truth** — all KPIs are computed by one metrics service, so every screen and export agrees.
- **Idempotency** — lead ingestion and billing writes are safe to repeat; duplicates are impossible.
- **Tenant isolation** — a portal user's client is derived from their token, never the request.
- **Automation over manual effort** — reminders, history and dormancy flags are computed server-side before every save.
- **Graceful degradation** — when an integration (Meta billing, AI) is unavailable, the affected panel degrades without breaking the page.

6.1 Architecture (functional view)

Layer	Responsibility
Front-end SPA	React single-page app; two shells (agency console, client portal); Redux state; lazy-loaded routes; offline curtain; auto-refresh timers.
API layer	Express REST API under <code>/api</code> ; JWT auth (header or cookie); per-route role/permission & tenant guards; centralised error handler; two-tier rate limiting.
Service layer	Meta & Google ad services, single metrics service, billing reconcilers, telecaller automation, Excel engine, AI client.
Background jobs	Cron scheduler (Google + Meta sync, staggered), webhook receiver, retry worker, zombie-run cleanup.
Data store	MongoDB (Cosmos DB) — 24 collections; unique/compound indexes enforce idempotency and integrity.
External systems	Meta Graph API, Google Ads API, Google Gemini.

6.2 The two experiences

Experience	Entry & token	Scope
Agency Console	<code>/login</code> · <code>token</code> / <code>user</code>	All clients; capabilities gated by staff role + permissions.
Client Portal	<code>/client-login</code> · <code>clientToken</code>	One client only; capabilities gated by portal role (admin / telecaller) + strict tenant scope.

6.3 Environment & conventions

- **Currency:** INR (₹). **Business day:** IST (Asia/Kolkata) for all daily roll-ups.
- **Response shape:** JSON `{ success: boolean, data|error|message }`.
- **Auth transport:** `Authorization: Bearer <jwt>` or secure `httpOnly token cookie ( SameSite=None; Secure )`.
- **Session default:** access token 30 days; refresh token 90 days (issued; used long-lived access token in practice).

7 Module Overview

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Twenty-five functional modules span the two experiences. Detailed behaviour follows in Sections 13–26.

ID	Module	Realm	Functional summary
M01	Authentication & Session	Both	Login, token issue/verify, logout, profile & password self-service, rate limiting, 401 grace handling.
M02	Access Management	Agency	Staff CRUD, role assignment, page/API permission grants, activate/deactivate, password reset.
M03	Client Management	Agency	Client CRUD, onboard, drop/re-onboard, cascade delete, search, staffing.
M04	Client Portal Provisioning	Agency	Enable portal, seed admin login, reveal/reset password, copy login URL.
M05	Google Ads Integration	Agency	Connect customer ID, validate, sync campaigns/metrics/keywords, associate/bulk-associate.
M06	Meta Ads Integration	Agency	Connect ad account & page, sync entities/insights/forms, webhooks, retry queue, form assignment.
M07	Lead Management	Both	Capture, attribute, assign, status, inline edit, manual create, delete-guard.
M08	Telecalling Sheet & Automation	Both	26-column sheet, auto-save, follow-ups, reminder/history/DARMANT automation, Excel import/export.
M09	Daily Lead Data / Daily Entry	Agency	Per-day per-client capture, auto totals/CPL, By-Client & Lead-Check views.
M10	Billing & Payments	Agency	Record payment (credit), spend debit, reconcile, adjust, reset, ledger.
M11	Balance Monitoring & Alerts	Both	Live balances, low-balance bell, Balance Watch buckets.
M12	Agency Dashboard	Agency	Cross-client spend/leads/CPL/balance, smart alerts, ranking, drill-through.
M13	Client Ad Details	Agency	Per-client Google/Meta/EOD tabs, KPIs, charts, leads, resync, AI insights.
M14	Ads Comparison	Agency	Multi-client side-by-side Google & Meta tables, rankings.
M15	Analytics & Reports	Both	Trends, per-client report, billing ledger, spend overview, staff/campaign/lead reports.
M16	Total Leads Pivot	Agency	Month-tabbed leads-by-client-by-day pivot, export.
M17	EOD Telecalling Report	Both	Day/month target-vs-achieved, appointment pipeline, day summary, editable targets.
M18	Monthly Abstract	Both	Per-day grid, banded columns, editable override cells.
M19	Client Portal — Today	Portal	Live KPI tiles + source cards, clickable drill-through, 30s refresh.
M20	Portal User Management	Portal	Create/edit/deactivate portal users, last-admin & self guards.
M21	AI Campaign Insights	Agency	Gemini analysis: summary, A–F grade, strengths, strategies, red-flags.
M22	Client Credential Vault	Agency	Encrypted client platform credentials, reveal/copy.
M23	Personal Vault	Agency	Personal encrypted credentials with sharing & auto-permission grant.
M24	Content Calendar	Agency	Plan/track social posts, calendar & list views, approval status.
M25	Sync Observability	Agency	Sync-run history, retry queue, raw-lead audit, health.

Every routed screen, its path, realm, access, and primary function.

Agency Console

Screen	Route	Access	Function
Login	/login	Public	Staff email/password sign-in.
Dashboard	/dashboard	dashboard	Performance command centre across all clients.
Leads Check (Daily Lead Data)	/daily-lead-data	daily-lead-data	Day-by-day per-client lead & spend; By-Client / Lead-Check views.
Clients	/clients	clients	Client master; add/edit, connect ads, drop/delete.
Total Leads	/leads	leads	Month-tabbed leads pivot by client & day.
Ads Comparison	/ads-dashboard	ads-dashboard	Multi-client Google/Meta comparison.
Client Ad Details	/client-ads/:clientId	clients / ads-dashboard	Single-client deep dive (Google, Meta, EOD).
Analytics / Reports	/reports , /analytics	reports	Trends, per-client analytics, exports.
Settings	/settings	Authenticated	Profile, avatar, password change.
Access Management	/access-management	admin/superadmin	Staff & permission administration.
Client Portal Access	/client-portal-access	admin/superadmin	Portal provisioning per client.
Daily Entry	/daily-entry	daily-entry	Manual daily marketing entry (secondary).
Content Management	/content-management	content-management	Social content calendar (hidden from nav).
Client / Personal Vault	/client-vault , /personal-vault	client-vault / personal-vault	Credential vaults (hidden from nav).
Campaign builders (mock)	/campaigns/*	Authenticated	Meta/Google/Email builder scaffolding — non-live.

Client Portal

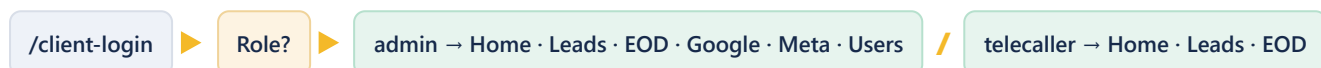
Screen	Route	Access	Function
Client Login	/client-login	Public	Username/email + password sign-in.
Portal Dashboard (shell)	/client-portal	All portal users	Role-gated tabs: Home, Leads, EOD, Google, Meta, Users.
Portal — Today	/client-portal (Home)	All portal users	Live today KPIs + source cards.
Portal — Leads (telecalling sheet)	/client-portal/leads	All portal users	26-column telecalling worksheet.
Portal — Google Ads	tab	Client admin	Own Google performance.
Portal — Meta Ads	tab	Client admin	Own Meta billing + leads.
Portal — Users	tab	Client admin	Manage portal users.

9.1 Agency console navigation



The sidebar renders only pages the user's permissions allow. `ProtectedRoute` validates the session and page permission on every navigation; unauthorised access redirects to the first permitted page. Deep links (Dashboard client card, Clients row) route to `/client-ads/:clientId`.

9.2 Client portal navigation



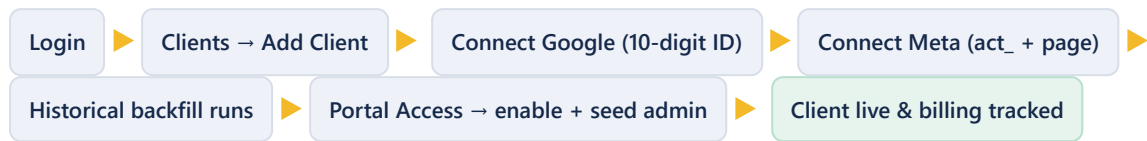
Telecaller-restricted tabs are both hidden and defended by redirect if reached directly. The portal shell carries a collapsible sidebar, client name and Logout.

9.3 Cross-cutting navigation behaviour

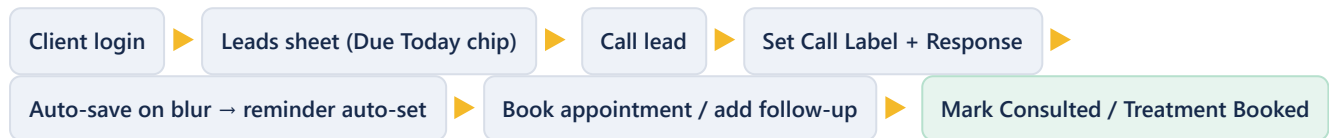
Behaviour	Detail
Lazy loading	Every page is code-split and prefetched after boot for instant subsequent navigation.
Offline curtain	Loss of connectivity shows a full-screen OfflineScreen; restores automatically.
Deep-link filters	Date range, client and month selections are URL-encoded so views are shareable/bookmarkable.
Drill-through	Dashboard tiles & source cards, and report figures, link to the Leads view pre-filtered by the clicked dimension.
Session expiry	A clearly token-related 401 (outside auth endpoints), after an 800 ms grace window with no intervening success, clears the session and returns to <code>/login</code> .

10 User Journey

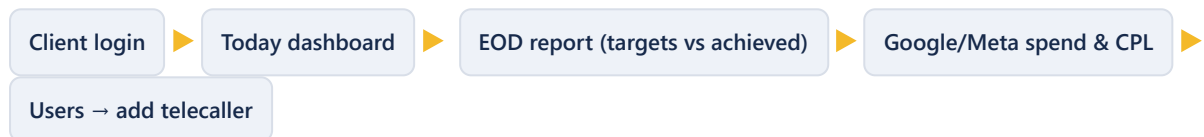
10.1 Account Manager — onboard a client



10.2 Telecaller — work the daily queue



10.3 Client Admin — verify results



10.4 Performance Manager — optimise



11 User Roles

Two independent role systems across two authentication realms.

11.1 Agency staff roles

`User.role` is free-form; only `superadmin` and `admin` are hard-coded full-access. All other roles are permission-driven.

Role	Nature	Default functional capability
superadmin	Root (protected)	All permissions; only role that can manage superadmins; cannot be deleted or assigned from UI.
admin	Full access	All pages & permissions; user admin, billing, all clients.
PMM	Performance marketing	Lead C/R/U, client C/R/U, fund C/R/U, entries, reports view+export.
SMM (default)	Servicing & content	Lead C/R/U, client R, fund R, entries, content C/R/U, report view.
staff (legacy)	Limited	Lead C/R/U, client R, fund R, entry R, report view.
Custom	As granted	Any mix of page + API permissions.

11.2 Client-portal roles

Role	Nature	Functional capability
admin	Full portal (own client)	Google, Meta, billing, leads/telecalling, EOD/Monthly, Users tab.
telecaller (default)	Lead handling	Today, Leads sheet, EOD. Ad spend/billing hidden.

Least-privilege default. Legacy portal tokens lacking a user identity are degraded to `telecaller` on protected portal routes (closing a prior escalation gap); on shared admin/portal routes the effective default is treated as `admin` for backward compatibility — both derive the client strictly from the token.

12 Permission Matrix

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12.1 Agency API permissions by role (defaults)

Permission	superadmin	admin	PMM	SMM	staff
user:* (create/read/update/delete)	✓	✓	—	—	—
client:create	✓	✓	✓	—	—
client:read	✓	✓	✓	✓	✓
client:update	✓	✓	✓	—	—
client:delete	✓	✓	—	—	—
lead:create / read / update	✓	✓	✓	✓	✓
lead:delete	✓	✓	—	—	—
entry:create / read / update	✓	✓	✓	✓	R only
fund:* (create/read/update)	✓	✓	✓	R only	R only
content:create / read / update	✓	✓	R only	✓	—
report:view	✓	✓	✓	✓	✓
report:export	✓	✓	✓	—	—
vault:read / update	✓	✓	—	—	—
settings:update	✓	✓	self	self	self

Enforcement order: (1) superadmin/admin bypass → allow; (2) explicit API permission present → allow; (3) holds a sidebar-page permission that maps to the API permission → allow; else **403**. Page→API mapping example: holding page **daily-entry** or **daily-lead-data** grants **entry:create**.

12.2 Portal role capability matrix

Capability	Client Admin	Telecaller
Today dashboard	✓	✓
Leads / telecalling sheet	✓	✓
EOD & Monthly reports	✓	✓
Google Ads / Meta Ads / billing	✓	✗ hidden
Manage portal users	✓	✗
Cross-client data	✗ blocked	✗ blocked

13 Functional Requirements

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Numbered functional requirements (FR) describing what the system does. Priority: **Must** **Should** **Could**.

ID	Functional Requirement	Pri.
FR-01	Authenticate agency staff (email+password) and portal users (username/email+password), issuing a scoped JWT.	Must
FR-02	Enforce agency RBAC (role bypass → API permission → page-permission mapping) and portal RBAC (admin/telecaller) with tenant scope.	Must
FR-03	Administer staff users: create (auto USRnnn), edit, toggle active, reset password, grant permissions; protect superadmin.	Must
FR-04	Manage clients: create, update (billing fields stripped), list (drop-hidden by default), soft-drop, re-onboard, cascade delete, text search.	Must
FR-05	Provision the client portal: enable, seed admin email/password, reveal saved password, copy login URL.	Must
FR-06	Connect & sync Google Ads: validate 10-digit customer ID, associate, sync campaigns/metrics/keywords, background & manual sync.	Must
FR-07	Connect & sync Meta: attach ad account + page(s), sync campaigns/adsets/ads/insights/forms; assign unmapped forms.	Must
FR-08	Ingest Meta leads via signed webhook (ack ≤ seconds), poll safety-net, and retry queue with exponential backoff; idempotent on lead id.	Must
FR-09	Present a 26-column telecalling sheet with inline edit, auto-save, follow-ups, filters and queue chips.	Must
FR-10	Run telecaller automation on every save: reminder auto-suggest, history append, auto-stamps, DARMANT flip.	Must
FR-11	Import/export the telecalling workbook (xlsx) with value cleaning, contact upsert, and history preservation.	Must
FR-12	Create manual leads (WhatsApp/walk-in) and block deletion of synced Meta-form leads.	Must
FR-13	Maintain an immutable, idempotent billing ledger: payment credit, spend debit (delta), reconcile, adjust, reset.	Must
FR-14	Track live ad-account balances and raise low-balance alerts (bell + Balance Watch).	Must
FR-15	Render the agency dashboard, Ads Comparison, per-client analytics, Total Leads pivot and Daily Lead Data views.	Must
FR-16	Render the EOD report (day/month targets, appointment pipeline) with editable per-client targets, and the Monthly Abstract with override cells.	Must
FR-17	Serve the client portal Today dashboard with clickable drill-through and 30 s auto-refresh.	Must
FR-18	Compute all KPIs via one metrics service; expose an ingestion-gap diagnostic.	Must
FR-19	Export reports & leads to Excel and PDF.	Must
FR-20	Generate AI campaign insights on demand with graceful degradation.	Could
FR-21	Store client & personal credentials encrypted; support personal-vault sharing with auto page-permission grant.	Should
FR-22	Record sync-run history, retry queue and raw-lead audit for observability.	Should

Behavioural specification of the principal screens (layout, controls, actions, states).

14.1 Login (staff) & Client Login

Aspect	Specification
Layout	Split-screen: brand poster + credential card.
Fields	Staff: Email, Password. Portal: Username or Email, Password (with show/hide toggle).
Actions	Sign In (submit). On success: persist token/user (or clientToken/clientData), route to first permitted page / portal home.
Failure	Invalid credentials → inline error; account inactive → blocked with message; login endpoints never force-redirect on 401. Login is rate-limited (10 / 15 min, successful attempts reset the counter).
States	Idle · Submitting (button spinner) · Error · Success.

14.2 Clients

Aspect	Specification
List	Paginated table; count tiles (Total/Active/Inactive); full-text search (name, place, industry, account ID). Dropped clients hidden unless <code>includeDropped=true</code> .
Row actions	View · Edit · Google-connect (G) · Meta-connect (f) · Drop/Delete. Integration icons turn green when linked.
Add/Edit dialog	Client Name (required) + Place, Industry, Address, GST, Onboard date, assigned SMM/SME, Team.
Google connect	Enter 10-digit Customer ID (+ optional name) → Connect & Launch → validates, associates, kicks background sync.
Meta connect	Step 1: Ad Account ID (<code>act_...</code>) → Save & Continue. Step 2: pick Facebook Page → Attach & Launch → begins ≤90-day backfill.
Drop	Requires reason; sets status <code>dropped</code> , records who/when, reversible via re-onboard.
Delete	Hard delete with confirmation; cascades across ~10 collections; reports per-collection outcome.

14.3 Telecalling Sheet (agency & portal)

Aspect	Specification
Grid	26 columns grouped: Lead Details · Initial Call · Reminder · Appointment · Follow-up · Action. Date-scoped (defaults today).
Queue chips	Due Today, Fresh, Call Back, etc., derived from reminder date & status.
Filter bar	Source, Call Label, Response, Hair/Skin, Appointment, Telecaller, free-text search; same-dropdown values OR, different dropdowns AND.
Inline edit	Cell edit → auto-save on blur (portal) / row Save button (agency); button shows Saving → Saved.
Row colour	Response tinting: green=Treatment Booked, sky=Consulted, purple=Duplicate; status chip palette per Section 15.
Follow-up	"+" adds a follow-up row (number, date, call label, remarks); FU number auto-increments; removable.
Manual lead	Add WhatsApp/Walk-in dialog: Name + Phone required; call/appointment fields optional.
Delete	Trash icon on manual leads only; confirmation dialog; synced Meta-form leads have no delete.
Import/Export	Excel import (upsert by contact, cleans values, preserves history, reports dirty/skipped); Excel & PDF export.

14.4 Access Management

Aspect	Specification
Create/Edit user	Name, Email, Role, Password; permission picker mirroring CONFIDENTIAL with role presets; admin-only page markers.
Actions	Toggle active, reset password, delete; superadmin cannot be assigned/deleted; only superadmin edits superadmin.

14.5 EOD Report & Monthly Abstract

Aspect	Specification
EOD sections	Header · Day Target-vs-Achieved · Month Target-vs-Achieved (+Projection) · Appointment Status (yesterday→+5d) · Day Summary (Leads Abstract, Total Calls, Connected, Appointments, Consultation/Conversion).
Editable targets	Click a Target cell → type → Enter; saved per client; percentages recompute live.
Monthly Abstract	One row per date + month TOTAL; banded columns (Source, Status, Calls, Appointments/Conversion); editable override cells auto-save per cell.
Refresh	Both silently re-fetch every ~30 s with a small loading indicator.

15 Field Definitions

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Definitions for the principal data fields. Enumerations reflect the exact system vocabulary.

15.1 Client (key fields)

Field	Type	Definition
clientName	String (req)	Client business name.
place / organisationType / address / gstNumber	String	Location, industry, address, GST number.
status	Enum	<code>active</code> · <code>inactive</code> · <code>pending</code> · <code>suspended</code> · <code>dropped</code> (dropped = soft-delete).
google_ads_customer_id	String (unique, sparse)	Exactly 10 digits.
meta_ad_account_id	String (unique, sparse)	Pattern <code>act_<digits></code> .
billing.*	Object	<code>billing_type</code> (monthly/quarterly), <code>total_added_funds</code> , <code>total_spend</code> , <code>available_balance</code> , <code>low_balance_threshold</code> (default 100).
telecalling_targets.*	Object	<code>day_consult</code> (10), <code>day_calls</code> (100), <code>month_consult</code> (310), <code>month_calls</code> (3100).
portalEmail / portalEnabled	String / Bool	Seed portal login & enable flag.

15.2 Lead — telecalling sheet columns (26)

Col	Field	Definition / values
1	date	Sheet DATE (lead day).
2	source_sheet	<code>WHATS APP</code> , <code>INSTAGRAM</code> , <code>FACEBOOK</code> , <code>GOOGLE LEAD</code> , <code>JUSTDIAL</code> , <code>WALK-IN</code> , <code>REFERRAL</code> , <code>PHYSICAL MARKETING</code> , <code>INCALL GOOGLE/FB/INSTA/SELF</code> .
4	contact	10-digit phone (unique within client).
5	lead_location	Location free-text.
6	hair_or_skin	<code>HAIR</code> · <code>SKIN</code> .
7	telecaller_name	Assigned telecaller.
8	first_call_date	Auto-stamped on first call-label set.
9	first_call_label	<code>CONNECTED</code> , <code>INVALID</code> , <code>BUSY</code> , <code>SWITCH OFF</code> , <code>WRONG NUMBER</code> , <code>RNA</code> , <code>DUPLICATE</code> , <code>DISCONNECTED</code> , <code>NOT CONNECTED</code> .
10	response_label	<code>HOT</code> , <code>WARM</code> , <code>COLD</code> , <code>DARMANT</code> , <code>CLOSED</code> , <code>APPOINTMENT BOOKED</code> , <code>REPEAT</code> , <code>INVALID</code> .
11	remarks	Call remarks.
12	reminder_date	Virtual alias of <code>next_followup_date</code> .
13	appointment_status	<code>APPOINTMENT BOOKED</code> , <code>...-VISITED</code> , <code>...-NOT VISITED</code> , <code>APPOINTMENT RESCHEDULED</code> .
14/15	appointment_date / appointment_booked_date	Appointment day; booked-on auto-stamped.
16	follow_up_number	Virtual = <code>follow_ups.length</code> .
17–19	follow_ups[]	Each: number, date, call_label (<code>CONNECTED</code> , <code>BUSY</code> , <code>SWITCH OFF</code> , <code>RNA</code> , <code>DISCONNECTED</code> , <code>NOT CONNECTED</code>), remarks, connected.
20/21	history_not_connected / history_connected	Auto-appended, read-only: <code>Attempt {n}: {YYYY-MM-DD} {LABEL} – {remarks}</code> .

22	status_sheet	HOT, WARM, COLD, DARMANT, CLOSED, Follow-up .
23/24/25	consulted_date / treatment_booked_date / treatment_value	Consultation, treatment-booking dates & value. CONFIDENTIAL
26	month	Virtual month name from date.

15.3 Status chip palette (col 22 / queue)

Status	Colour intent
HOT	Red
WARM	Amber
COLD	Blue
CLOSED	Slate
DARMANT	Grey
Follow-up	Indigo

Validation rules enforced at the UI, API and schema layers. Enum fields are validated against the exact vocabulary in Section 15.

Field	Layer	Rule / on-failure behaviour
User.email	Schema	Required, unique, lowercase, must match email regex → "Please add a valid email".
User.password	Schema	Required, min length 6; bcrypt-hashed on save; never returned.
User.userID	Schema/hook	Auto-generated sequential <code>USRnnn</code> ; unique.
Client.clientName	Schema	Required → "Please add a client name".
Client.google_ads_customer_id	Schema	Optional; if present must be exactly 10 digits → "Google Ads Customer ID must be exactly 10 digits"; unique/sparse.
Client.meta_ad_account_id	Schema	Optional; must match <code>act_\d+</code> → "Meta Ad Account ID must start with \"act_\" followed by digits"; unique/sparse.
Client.billing.*	API	Stripped from generic client update — ledger-owned; cannot be set via update.
ClientPortalUser.username	Schema	Required, lowercase, 3–60 chars, pattern <code>^[a-z0-9._-]+\$</code> → "Username may only contain lowercase letters, digits, dot, underscore, hyphen"; unique per client.
ClientPortalUser.password	Schema	Required, min 6; bcrypt.
ClientPortalUser.role	Schema	Enum <code>admin · telecaller</code> .
ClientPortalUser.email	Schema	Optional; unique per client when non-empty (partial index).
Lead.contact	Schema/index	Unique within a client (partial, non-empty); duplicate → 400 "<field> already exists".
Lead.name / email	Schema	Required (email lowercased); manual create synthesises a placeholder email if absent for schema compliance.
Lead enum fields	API + Schema	call/response/status validated against enums by the controller (nice field-level error) before save; out-of-enum rejected.
AbstractEntry.date	Schema	Must match <code>^\d{4}-\d{2}-\d{2}\$</code> ; unique per (client, date).
Payment.amount	Schema	Required; ledger min 0.
Manual lead dialog	UI	Name and Phone mandatory before submit.

16.1 Request-layer validation messages (exact)

Enforced by the `validate` middleware; failure returns `400 { success:false, errors:[{field, message}] }`.

Field	Constraint	Exact message
name (register)	Required, 2–50 chars	"Name is required" / "Name must be between 2 and 50 characters"
email	Required, valid email	"Email is required" / "Please provide a valid email"
password	Required, min 6	"Password is required" / "Password must be at least 6 characters"
clientName	Required, 2–200 chars	"Client name is required" / "Client name must be between 2 and 200 characters"
lead.phone	Optional, mobile	"Please provide a valid phone number"
lead.source / status	Enum	"Invalid source" / "Invalid status"
dailyLeadData.date	Required, ISO8601	"Date is required" / "Invalid date format"
metaData.spend	Float ≥ 0	"Meta spend must be a non-negative number"

id (param)	Mongoid	"ID is required" / "Invalid ID format"
pagination limit	Int 1-10000	"Limit must be between 1 and 10000"

16.2 Client-side (UI) validation — exact messages

Screen · field	Rule	Exact message
Login · email/password	Yup email + min 6	"Invalid email address" · "Password must be at least 6 characters"
Clients · required set	Name, Place, Industry, Address required	"Client name is required" · "Place is required" · "Industry is required" · "Address is required"
Clients · Account ID	<code>/^\d+\$/</code>	"Account ID must contain only numbers"
Clients · Google Customer ID	<code>/^\d{8,12}\$/</code> (UI) — schema requires exactly 10	"Customer ID must be 8–12 digits (e.g. 1234567890 or 123-456-7890)."
Clients · Meta Ad Account	<code>/^act_\d+\$/</code>	"Ad Account ID must be in the format act_1234567890"
Portal user · username	<code>/^[a-z0-9._-]{3,60}\$/</code>	"Username: 3–60 chars, lowercase letters/digits/dot/underscore/hyphen only"
Manual lead · name/phone	Required	"Name is required" · "Phone is required"
Settings · password change	Current req · new ≥ 6 · confirm match	"New password must be at least 6 characters" · "New password and confirmation do not match"
Import · file type	<code>/\.xlsx\$/i</code>	"Please select an .xlsx file (Excel workbook)."

Layering: client-side checks give immediate feedback; the API re-validates (express-validator + controller enum checks); the schema is the final guard. Mongoose validation errors are flattened to a single 400 message joined by ", " (see Section 27). Note the deliberate looseness of the UI Google-ID rule (8–12 digits) versus the schema's strict 10-digit rule — the server is authoritative.

17 Workflow Definitions

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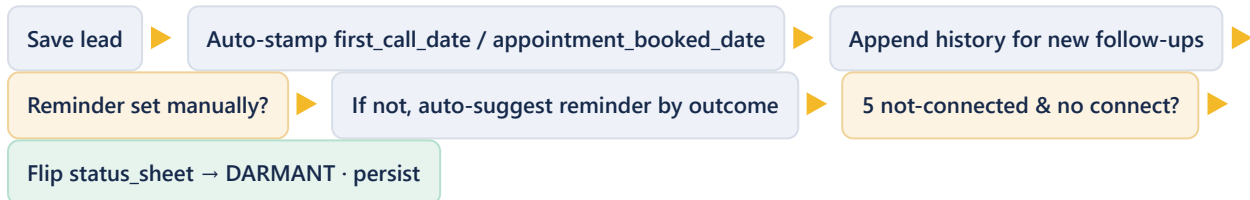
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17.1 Meta lead ingestion (UML activity — textual)

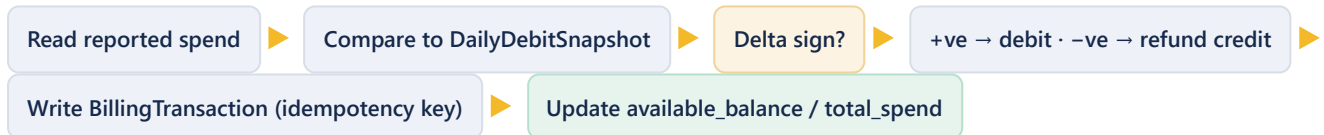


Invalid signature → 401, no processing. A poller independently re-checks each form for missed leads; the ingestion-gap diagnostic reports any shortfall vs Meta's reported counts.

17.2 Telecaller save automation (runs before persist)



17.3 Billing reconciliation (per client/campaign/day)



17.4 Client onboarding



18.1 Lead state machine

From	Trigger	To
New / Fresh	First call logged	Contacted
Contacted	Response HOT/WARM/COLD	HOT / WARM / COLD (reminder +2/+7/+30d)
Any active	Response = APPOINTMENT BOOKED + date	Appointment Booked (reminder = day before appt)
Appointment Booked	Appointment status update	Visited / Not Visited / Rescheduled
Any	Response = CONSULTED (consulted_date)	Consulted
Consulted	Response = TREATMENT BOOKED / CLOSED	Converted / Closed (reminder cleared)
Any	5 consecutive not-connected, no connect	DARMANT (auto, darmant_flagged=true)
DARMANT	Manual status change	Re-engaged (flag preserved unless un-flagged)
Any	CLOSED / INVALID / DUPLICATE / WRONG NUMBER	Terminal (reminder cleared)

18.2 Client lifecycle states

From	Trigger	To
—	Create	active
active	Status change	inactive / pending / suspended
active/any	Drop (with reason)	dropped (hidden, history retained)
dropped	Re-onboard	active (visibility restored)
any	Hard delete	Removed (cascade)

18.3 Meta sync-run states

State	Meaning
running	In progress (overlap-guarded; zombie if untouched 30+ min after crash).
success / partial / failed	Completed cleanly / with per-client errors / aborted.

18.4 Webhook-retry states

State	Meaning
pending	Awaiting next attempt (backoff schedule).
resolved	Lead successfully processed on retry.
abandoned	Exhausted retries (after 6h tier).

Rule	Definition
BR-01	Superadmin/admin bypass all agency permission checks.
BR-02	Portal user's client is derived from the token, never the request (tenant isolation).
BR-03	Meta lead processing is idempotent on <code>meta_leadgen_id</code> .
BR-04	Contact number unique within a client; may repeat across clients.
BR-05	Synced Meta-form leads cannot be deleted; manual leads can.
BR-06	DARMANT flips after 5 consecutive not-connected attempts with no connection.
BR-07	Reminder auto-set: BUSY→+1h; RNA→other half-day (AM→15:00, PM→next 10:00); SWITCH OFF/NOT CONNECTED/DISCONNECTED→tomorrow 10:00; HOT/WARM/COLD→+2/+7/+30d; APPOINTMENT BOOKED→day before appt; CLOSED/INVALID/DUPLICATE/WRONG NUMBER→clear.
BR-08	WhatsApp leads counted only on CONVERSATIONS-optimised ad sets.
BR-09	Reported lead counts come from Meta insights, not CRM records alone.
BR-10	Every ledger entry carries an idempotency key; duplicates roll back.
BR-11	Billing balances are ledger-owned; never editable via client update.
BR-12	Sync windows floored at client onboardDate.
BR-13	Meta billing runs dry-run by default until explicitly enabled.
BR-14	Dropped clients excluded from live lists; drop history retained.
BR-15	Last active portal admin cannot be deleted/demoted/deactivated; users cannot delete themselves.
BR-16	History line format is fixed: <code>Attempt {n}: {YYYY-MM-DD} {LABEL} – {remarks}</code> ; connected vs not-connected routed to separate columns.
BR-17	<code>first_call_date</code> stamped on first call label; <code>appointment_booked_date</code> stamped when <code>appointment_date</code> first set.
BR-18	All money INR; business day IST.
BR-19	Telecalling values validated against the exact spreadsheet vocabulary for clean Excel round-trip.

20 Notification Behaviour

The platform uses in-app alerts and live indicators rather than external push/email (see Section 36 for planned channels).

Notification	Trigger	Behaviour
Low-balance bell	Meta ad-account balance below threshold (e.g. under ₹2,000)	Header bell (admins) badges affected accounts; auto-refreshes ~90 s.
Balance Watch board	Balance buckets	Clients grouped Critical / Warning / Healthy on the dashboard.
Smart dashboard alerts	Heuristics	"Low balance — ads may auto-pause", "Spending but zero leads", etc.
Live-view dot	Auto-refresh cycle	Small green/loading dot indicates a live view refreshing (~30 s).
Save feedback	Row/cell save	Inline "Saving..." → "Saved" transient state.
Toasts / snackbars	CRUD success/failure	Success/error toast per action (create, update, delete, import).
Confirmation dialogs	Destructive actions	Delete client/user/lead, drop client, reset billing require explicit confirmation.
Meta billing unavailable	Token/permission error	Yellow inline notice ("Live billing not available — contact account manager"); rest of page functions.

21 Search Behaviour

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Context	Behaviour
Clients	Server-side full-text index over clientName, place, organisationType, accountID; instant filter as you type.
Leads / telecalling sheet	Free-text search matches name, email, phone, location, remarks, telecaller and every form answer.
Access Management	Search staff by name, email, userID.
Dashboard / Ads Comparison	Client-name search filters cards/rows client-side.
Content calendar	Filter by clientName, status, assignedSME.

22 Filtering Behaviour

Context	Behaviour
Telecalling filter bar	Multi-select Call Label, Response, plus Source, Hair/Skin, Appointment, Telecaller. Same dropdown = OR; different dropdowns = AND. "Showing X of Y" counter; Clear-all when any filter active.
Queue chips	One-click buckets (Due Today, Fresh, Call Back...) derived from reminder date & status.
Date range	Quick buttons (Today, 7, 14, 30 days) or custom From/To; URL-encoded; Daily/Weekly/Monthly toggles on Lead Check.
Client filter	Daily Lead Data, Analytics and reports filter to one client or All Clients.
Dropped clients	Hidden by default; management view opts in via <code>includeDropped</code> .
Range guard	Server caps certain pivots at ≤ 90 days to protect performance.

23 Sorting Behaviour

Context	Behaviour
Leads	Newest first by Meta form-submission time (meta_created_time) / createdAt; manual entries appear at top on add.
Clients	Default by name / creation; count tiles summarise status.
Dashboard rankings	Top Performing Clients ranked by today's leads; Performance Insights rank Top / Best-Value / Needs-Review.
Reports/pivots	Chronological by date; TOTAL row pinned (Monthly Abstract top row).
Data grids (MUI)	Column sort where the data grid is used (e.g. analytics tables).

24 Report Behaviour

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Report	Behaviour
EOD Telecalling Report	Per-day + month-to-date roll-up: source buckets, connected/not-connected/invalid-duplicate, fresh vs callback calls, appointment pipeline (yesterday→+5d), editable day/month targets with live projection %. Achieved cells highlight red on a miss. Auto-refresh ~30 s.
Monthly Abstract	One row per calendar date + month TOTAL; banded Source / Status / Calls / Appointments-Conversion columns; manual-override cells (allow-listed) auto-save per cell; spreadsheet-faithful styling.
Daily Lead Data	By-Client (one row per client/day: Meta form/WhatsApp/fund, Google call/website, spend) and Lead-Check rollup with Daily/Weekly/Monthly toggles; totals row; PDF export.
Total Leads pivot	Month-tabbed; one column per day; cell = total lead count (Meta form + WhatsApp + Google call + website); Excel/PDF export.
Per-client Analytics	Daily metrics, per-campaign rollups (KPIs from totals, not day-averages), impression-share averaging, top-N keywords, billing ledger with balance timeline; 5-min cache; first-ever call triggers a one-time blocking sync.
Reports module	Dashboard overview (client/user counts), campaign-performance (Meta vs Google ROI/CPL), lead-performance (by status/source, conversion rate); user-performance & financial reports gated to admin/superadmin (financial currently stubbed).
Exports	Excel (ExcelJS — full columns, form answers, split follow-up history; styled headers, frozen panes, duplicate tinting) and PDF (landscape A3 for leads).
Consistency	All lead/spend KPIs computed by the single metrics service; every report agrees.

25.1 Agency dashboard

Element	Behaviour
Summary cards	Total Clients, Active, Total Leads today, Total Spend today, Average CPL — date-scoped (defaults today, date picker top-right).
Client summary	Meta/Google/both-connected breakdown; cumulative Meta balance.
Overview tab	Balance Watch, Performance Insights, Alerts, Recent Activity.
Clients tab	Searchable client list with merged Meta+Google metrics; per-client card: META (Form, WhatsApp, Spend, CPL) + GOOGLE (CTR, CPC, Clicks, Impressions).
Ranking	Top Performing Clients by today's leads.
Drill-through	Click a client → <code>/client-ads/:clientId</code> .
Data source	Bulk dashboard-overview endpoint (replaced N+1); falls back to Meta insights leads when CRM records are absent.

25.2 Client portal — Today

Element	Behaviour
Six KPI tiles	Total Leads today, Connected, Total Calls, Appointments, Consulted, Conversions.
Six source cards	WhatsApp, Instagram, Facebook, Google Lead, Walk-in/Referral, Incall.
Activity boxes	Call Activity (fresh, callbacks, connected, not-connected, connect-rate bar); Appointments (today, booked today, upcoming 5 days).
Interactivity	Every tile/card clickable → Leads page filtered by that dimension.
Live	Auto-refresh ~30 s with a live indicator dot.

25.3 Client Ad Details (per client)

Tabs: **Google Ads** (account info, billing/utilisation, KPI cards, spend-trend chart, campaigns with expandable keywords), **Meta Ads** (billing, KPI grid — Spend, Total/Form/WhatsApp Leads, CPL, Reach, Impressions, Clicks, CTR, CPC, CPM — campaigns, lead forms, leads table, AI insights), **EOD Report**. Live Resync-from-Meta, 5-min caching, Excel/PDF lead export.

REST API under `/api`; JSON `{success, data|error|message}`; JWT-guarded. Representative behaviour by group.

Group	Guard	Key behaviour
<code>/auth</code>	Public / protect	login, client-login, register (admin), me, client-me, logout, update-details, update-password (re-issues token).
<code>/users</code>	admin/superadmin	Staff CRUD, stats, roles, teams, toggle-status, change-password, permissions; superadmin protections.
<code>/clients</code>	checkPermission	list (drop-hidden), create, update (billing stripped), delete (cascade), status, drop, reonboard, portal-credentials reveal.
<code>/leads</code>	checkPermission	CRUD, stats/summary, monthly-meta-by-client (60s cache), daily-by-client ($\leq 90d$, ingestion-gap diagnostic).
<code>/meta</code>	protectAdminOrClient	webhook (public, HMAC), sync (+historical), form assign/reprocess, per-client config/analytics/spend-overview, leads CRUD + import/export, telecalling-report, targets, monthly-abstract + cell; agency-only observability.
<code>/google-ads</code>	admin/superadmin	sync (all/one), campaigns, associate/bulk-associate, validate-customer-id, sync-status.
<code>/metrics</code> · <code>/analytics</code>	protectAdminOrClient	Google aggregates & per-client report + billing ledger; tenant-scoped; caches.
<code>/daily-entries</code> · <code>/daily-lead-data</code>	checkPermission	Daily capture, stats, date lookups; unique per (client,date); pre-save CPL calc.
<code>/payments</code> · <code>/billing</code>	admin/superadmin	record payment (credit + rollback), reconcile, deep-sync, reset (confirm-guarded, destructive).
<code>/client-portal</code>	protectClient	own analytics (empty shell for Meta-only, never 400), portal user CRUD (admin, last-admin guard).
<code>/reports</code>	report:view	dashboard, campaign, lead reports; user/financial gated admin.
<code>/content-entries</code> · <code>/vault</code> · <code>/personal-vault</code> · <code>/ai</code>	varies	Content CRUD; vault CRUD (decrypt on read); personal-vault with sharing; AI analyze-campaign.

Known control gap. `/api/vault` is guarded by authentication only (no role/permission gate) and returns decrypted client credentials to any authenticated staff user — flagged for remediation (Sections 29 & 36).

A centralised error middleware normalises all failures to `{ success:false, error:<message> }` with an appropriate status.

Condition	Status	Message / behaviour
Invalid Mongo ObjectId (CastError)	404	"Resource not found".
Duplicate key (11000)	400	"<field> already exists" (e.g. duplicate email, contact, customer/ad-account ID).
Schema ValidationError	400	All field messages joined by ", " (e.g. "Google Ads Customer ID must be exactly 10 digits").
Invalid JWT	401	"Invalid token".
Expired JWT	401	"Token expired".
Permission denied	403	"Access denied. Required permission: <perm>" / "Required role: ...".
Cross-tenant portal access	403	"Portal token cannot access another client".
Unknown route	404	"Not Found - <url>".
Rate limit exceeded	429	Login: "Too many login attempts, please try again later." · API: "Too many requests from this IP, please try again later."
Unhandled server error	500	"Server Error" (stack included only in development).
Meta webhook bad signature	401	Rejected before processing.

Async handlers are wrapped so rejected promises route to the middleware rather than crashing the request. Development mode includes the stack trace; production suppresses it.

28 Exception Handling

Scenario	Handling
Integration auth/permission failure	Typed errors (MetaAuthError, MetaPermissionError...) let callers stop or skip a client; UI shows a graceful "billing unavailable" notice; sync continues for other clients.
Rate limiting (Meta/Google)	Header parsing detects $\geq 80\%$ usage; sequential sync, staggered cadence, exponential backoff (2s/8s/32s); overlap guard skips a tick if the prior run is active.
Transient network error	Retried with backoff; cursor pagination capped ($\leq 10,000$ /edge).
Webhook processing failure	Enqueued to MetaWebhookRetry with backoff; poller safety-net; abandoned after final tier.
Process crash mid-sync	Zombie "running" sync-runs (untouched 30+ min) auto-marked failed at boot before the scheduler restarts.
Concurrent ledger writes	Idempotency-key duplicate is caught; the cached balance change rolls back so the net effect is one entry.
Payment recording failure	Full rollback of balance + ledger on error (idempotency key <code>payment:<id></code>).
AI service down / key missing	Graceful 503; feature degrades; truncated-JSON responses are repaired before use.
Encryption key missing	Portal password login still works (bcrypt); recoverable-copy feature disabled with a logged warning.
Unhandled rejection / uncaught exception	Logged; server closes gracefully / exits so the orchestrator restarts a clean process.

Edge case	Defined behaviour
Same lead via webhook + poll	Idempotent upsert on leadgen_id → exactly one Lead.
Same phone across two clients	Allowed (unique only within a client via partial index).
Blank contact numbers	Partial unique index excludes empty strings — no false duplicate collisions.
Lead form not yet mapped to a client	Lead deferred as "unmapped"; surfaces in admin form-assign queue.
Meta reports leads but CRM has none	Dashboard falls back to Meta-insights counts; ingestion-gap diagnostic flags the shortfall.
Page-only client (no ad account)	Ad-account sections hidden; form/lead sync still runs.
Client never synced opens analytics	First call triggers a one-time blocking sync before returning data.
Meta-only client opens portal Google analytics	Returns an empty shell (never 400).
Legacy portal login (no ClientPortalUser)	Falls back to legacy portalEmail/Password and auto-seeds an admin user.
Legacy portal token (no userId)	Degraded to least-privilege telecaller on protected portal routes.
Delete last portal admin	Blocked with guard message.
Delete synced Meta lead	Prevented (no delete affordance / API refusal).
Re-onboard a dropped client	Restores to active with full prior history.
Reminder rule conflict (multiple labels)	Precedence: appointment-with-date > response bucket > call-label bucket; terminal clears win.
Idempotent PUT resending unchanged follow-ups	History not re-appended (guarded by prev-length comparison).
Dirty Excel import values (RNR, DARMONT, WRONG NO)	Cleaned to canonical values; per-row "dirty fixes" and "skipped" reported.
Transient single 401 amid healthy traffic	800 ms grace window; cancelled by any intervening 200 — no session kill.
Meta ad-account balance unreachable	Yellow "live billing not available" notice; rest of page works.
Billing reset	Requires <code>confirm:true</code> ; wipes ledger/snapshots and sets a new opening balance.

Representative machine/system-level messages returned by the API and infrastructure.

Message	Context
"Resource not found"	Bad ObjectId (404).
"<field> already exists"	Duplicate-key on unique field (400).
"Invalid token" / "Token expired"	JWT verification failure (401).
"Access denied. Required permission: <perm>"	Missing agency permission (403).
"Access denied. Required role: <roles>"	Role gate failure (403).
"Portal token cannot access another client"	Tenant-scope violation (403).
"Not Found - <url>"	Unknown route (404).
"Too many login attempts, please try again later."	Auth rate limit (429).
"Too many requests from this IP, please try again later."	API rate limit (429).
"Server Error"	Unhandled 500 (stack in dev only).
Response envelope	Success: { success:true, data message }; failure: { success:false, error }.
Boot log: "cleaned up N orphaned MetaSyncRun records"	Zombie sync-run cleanup at startup.
Health: GET /health	Returns { success, uptime, timestamp }.

User-facing messages by interaction type (exact wording may be refined in the UI copy).

Type	Examples / behaviour
Login errors	Invalid credentials, inactive account → inline error under the form; no forced redirect on login 401.
Save feedback	Row/cell: "Saving..." → "Saved" (green); toast on create/update success and on failure.
Create/Update success	Client / user / lead / payment created or updated → success toast; list refreshes.
Delete confirmations	Client delete (shows cascade impact), user delete, manual lead delete (shows name/phone/location/category), billing reset — all require explicit confirm.
Portal user guards	Cannot delete/deactivate self; cannot remove the last active admin — blocked with an explanatory message.
Import result	Summary of rows imported, per-row dirty fixes applied, and rows skipped.
Integration notices	"Live billing not available — contact the account manager" (Meta token issue); "First sync may take a few minutes" guidance after connecting.
Empty states	No leads / no clients / not-linked banners with a call-to-action (e.g. "Link from the Clients page").
Offline	Full-screen offline curtain until connectivity returns.
Validation hints	Required-field markers; username rule hint; 10-digit / act_ format hints on connect dialogs.

31.1 Representative exact strings

Screen	Verbatim message
Clients (success)	"Client added successfully!" · "Client updated successfully!" · "{name} dropped — record kept for re-onboarding." · "{name} re-onboarded — back on the active list." · "{name} deleted permanently."
Clients (Meta connect)	"{name} connected. Meta is fetching up to 90 days of campaign, insights, and lead data in the background — this can take several minutes..."
Clients (delete confirm)	"This cannot be undone. Every lead, daily entry, fund entry, content entry, and vault record linked to this client will be deleted... Use Drop instead if there's any chance you'll re-onboard them later."
Access Management	"User created successfully" · "User deleted successfully" · "Admin users automatically get access to every page. No selection needed."
Portal Users	"Cannot demote or deactivate the last active admin. Promote another user first." · "You cannot delete your own account"
Import result	"{n} inserted" · "{n} updated" · "{n} value fixes" · "{n} skipped" · "Row {r} skipped — {reason}"
Settings	"Profile saved." · "Password updated." · strength meter: Too short / Weak / Fair / Good / Strong
Offline	"You're offline" · "Leadmatrix couldn't reach the internet. We'll retry automatically the moment your connection is back."
Login progress	"Signing in..." → "Sign in"

Consistency note for engineering. Failure payload keys vary across the codebase: most controllers return `{ success:false, message }`; express-validator returns `{ success:false, errors:[...] }`; the central handler and the payments/billing routes return `{ ..., error }`. Duplicate-key conflicts return 409 in portal-user creation but 400 in staff-user creation. 422 is never emitted. Standardising these is a recommended enhancement.

32.1 Meta Graph API

Aspect	Behaviour
Auth	Agency system-user token for ad reads; per-page tokens (encrypted at rest) for form/lead reads.
Sync	Cron every ~15 min (offset +5 from Google); startup run; overlap guard; sequential per client; version-pinned; explicit field sets (never <code>fields=*</code>).
Entities	Campaigns, ad sets, ads, insights (campaign/adset/ad, per day, rolling ~90d), lead forms, leads.
Webhooks	<code>GET</code> verify (hub.challenge); <code>POST</code> HMAC-SHA256 verified against raw body; ACK 200 then async process.
Billing	Spend pass-through; delta reconcile per campaign/day; idempotency key <code>meta:<client>:<campaign>:<date>:<cost></code> ; dry-run default.
Balance	Live available balance scraped from funding-source details; cached for fast dashboard reads.

32.2 Google Ads API

Aspect	Behaviour
Auth	OAuth refresh token + developer token + MCC login-customer-id.
Sync	Cron every ~15 min; campaigns, per-day metrics (incl. impression-share, click-type breakdown), keyword metrics; window floored at onboardDate (regular 30d, deep = onboardDate→today).
Billing	Same delta-reconcile model; ledger sources <code>google_ads_daily_spend/refund</code> .
On-demand	Analytics first-open triggers a one-time blocking sync; new customer-ID association kicks a background sync.

32.3 Google Gemini (AI)

Aspect	Behaviour
Endpoint	<code>POST /api/ai/analyze-campaign</code> (agency auth). Sends platform, client, date range, summary, top campaigns.
Output	Strict JSON: summary, A–F grade, strengths, improvements, 5 strategies, red-flags, KPIs to watch.
Robustness	Low temperature, token cap, 30s timeout, truncated-JSON repair, normalised output; missing key → 503.

32.4 Companion services

The Daily-Entry module additionally calls an external "Main API" (Azure-hosted) for clients/leads/funds and a separate Meta-sync API, with short-TTL caching — indicating a companion/legacy microservice topology alongside the in-process sync.

33 Audit Logging Behaviour

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Audit surface	What is captured
MetaLeadRaw	Immutable record of every raw Meta lead payload (source: webhook/poll/manual), received time, processed flag, resulting lead id, error — enables replay & idempotency.
MetaSyncRun	Each sync execution: scope, timing, per-entity counts (campaigns/adsets/ads/insights/forms/leads), per-stage errors, rate-limit usage, status.
MetaWebhookRetry	Dead-letter queue with attempts, next_attempt_at, last_error, status.
BillingTransaction	Immutable ledger of every credit/debit/adjustment/refund with balance_after, source, reference (payment/campaign/date), idempotency key, occurred_at/recorded_at.
DailyDebitSnapshot	Per client/platform/campaign/day debited vs reported amounts and last_synced_at.
Client drop history	drop_history[] of drop/reonboard actions with reason, timestamp and actor.
User activity	lastLogin stamp; createdBy/recordedBy/assignedTo attribution on leads, entries, content, payments.
Request logging	HTTP access logs (morgan) — combined in production, dev format otherwise.
Diagnostic logs	Tagged, password-free login diagnostics ([client-login]), sync stage logs, orphan-cleanup logs.

Note. Financial and lead-ingestion audit trails are immutable and idempotency-keyed by design; the system favours append-only records for anything money- or lead-related. A general end-user action audit log (per-field change history) is a candidate enhancement (Section 36).

34.1 External

Dependency	Reliance
Meta Graph API	Ad data, webhooks, page tokens; rate limits & version currency.
Google Ads API	Campaign/metric/keyword data via OAuth + developer token + MCC.
Google Gemini	AI insights (advisory).
MongoDB (Azure Cosmos DB)	Primary datastore.
Facebook Business Manager access	Client must grant agency access before Meta setup succeeds.
Azure hosting	App + scheduler runtime, HTTPS.

34.2 Internal / technical

Component	Role
Node.js + Express	API & scheduler.
React + Redux Toolkit + MUI	Front-end (both realms).
Cron scheduler + retry worker	Background sync & resilience.
Single metrics service	Canonical KPI definitions.
bcrypt + AES encryption utility	Password & secret protection.
ExcelJS	Telecalling workbook import/export.
Shared enum constants	Frontend/back-end vocabulary lockstep for clean round-trip.

35 Acceptance Criteria

Given–When–Then acceptance criteria mapped to functional requirements.

Maps to	Criterion
FR-02	Given a portal user, when they request another client's data, then the API returns 403 "Portal token cannot access another client" and no data.
FR-03	Given a new staff user, when created, then a unique <code>USRnnn</code> is assigned and the password is stored hashed and never returned.
FR-04	Given a client update including billing fields, when submitted, then billing fields are ignored and only permitted fields change.
FR-06	Given a customer ID that is not 10 digits, when saved, then the system rejects it with "Google Ads Customer ID must be exactly 10 digits".
FR-08	Given the same Meta lead delivered twice, when processed, then exactly one Lead exists and the raw payloads are both audited.
FR-10	Given a fifth consecutive not-connected attempt with no prior connection, when saved, then <code>status_sheet</code> becomes DARMANT and <code>darmant_flagged</code> is true.
FR-10	Given a response of WARM, when saved without a manual reminder, then <code>next_followup_date</code> is set to +7 days.
FR-11	Given an Excel import with "RNR" and "DARMONT", when imported, then values normalise to "RNA"/"DARMANT", rows upsert by contact, history is preserved, and fixes are reported.
FR-12	Given a synced Meta-form lead, when deletion is attempted, then it is blocked; a manual lead deletes after confirmation.
FR-13	Given a day's campaign spend, when reconciliation runs multiple times, then the balance is debited exactly once for that amount.
FR-14	Given a balance below threshold, when staff view the console, then the bell and Balance Watch flag it.
FR-18	Given a client's lead count, when viewed on dashboard, analytics and export, then all three agree.
FR-20	Given the AI key is absent, when insights are requested, then the system returns 503 and the page remains usable.
FR-01	Given 10 failed logins in 15 minutes, when another is attempted, then a 429 rate-limit message is returned.

36 Future Functional Enhancements

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Enhancement	Functional description
Vault access control	Add role/permission gating and access auditing to the client credential vault (closes the control gap in Section 26).
End-user change audit log	Per-field change history for leads and clients (who changed what, when) beyond the current financial/lead audit surfaces.
External notifications	Email/WhatsApp/SMS for low balance, appointment reminders, dormant-lead nudges and daily summaries.
Refresh-token rotation	Wire a refresh-exchange endpoint for shorter access tokens with seamless renewal.
Live Meta billing	Graduate Meta billing from dry-run to live after validation.
Restore financial reports	Reconnect stubbed financial/sales reports to a replacement fund data source.
Payment gateway	Optional online payment intake that auto-credits the ledger.
Telephony integration	Click-to-call and automatic call-outcome capture into the sheet.
Content & campaign modules	Promote the content calendar and campaign builders from scaffolding to live, data-backed features.
Deeper AI advisory	Cross-client, trend-aware recommendations and anomaly detection.
Additional channels	Extend integration to further ad platforms (YouTube, LinkedIn).
Native mobile app	Telecaller-focused mobile experience.

End of Document. This Functional Specification Document describes the ARA Lead Matrix CRM as built and in production as of 17 July 2026. It is issued as version 1.0 (baseline), companion to the Business Requirements Document ARA-BRD-CRM-2026-001, and is classified Confidential.